NATHAZ AI LAB / TECHNICAL REPORT · EXPLORATORY TRIAL

R23.4b Convection-scale AI and thunderstorm nowcasting

Group AI Weather Models

Status Closed (negative result)

Period 2026-10-14 14:21 — 2026-10-27 21:25

Stage conclusion

Project report

Convective-scale AI and nowcasting (sub-projects L / M / L2)

Objective

Determine whether ~3 km convective-scale AI models (StormCast, WoFSCast, etc.) directly predict thunderstorms and surface gusts. Assess whether they can run forecasts or hindcasts in batches of dozens to hundreds of events, which the PI identified on 10-14 as the only scalable thunderstorm pathway. Step 1 deliverables cover three items. First, compile a verification table testing whether public models run on a Colab A100 GPU. Second, build a verification table comparing 0–6 h lead-time forecasts against Automated Surface Observing System (ASOS) observations and High-Resolution Rapid Refresh (HRRR) baselines across 5–10 observed downburst or severe thunderstorm events. Third, compare AI and HRRR soundings against observed soundings for those same events as the prerequisite gate for sub-project L2.

Current status

10-28 lead re-review (6/6, consistent with prior conclusions; this round was a duplicate submission that the lead had not noticed at the time): the failed verdict and the negative result (limited to this configuration) stand. Documentation updates record that the numerical rule for "false alarms show no discriminating power" was set post hoc (deviation D8). Report-only metrics show any-member hits at 33/120 versus 33/120 and a member-maximum peak ratio of 1.096. Day-cluster intervals are not paired intervals, and the 35 dBZ check covers the event side only. Finalized; decision recorded. Exploratory; not peer reviewed.

10-27 S2 conclusion finalized (verified by the lead: primary criteria match digit for digit). Report-only metrics with day-cluster intervals using the HRRRCast heuristic show a Brier score (BS) difference of −0.028 [−0.073,0.016] and a hit-rate difference of +0.108 [0.023,0.197]. The peak ratio reaches 1.161, while only the false-alarm difference of +0.100 [0.048,0.156] fails. Exploratory; not peer reviewed.

10-27 21:45 six-reviewer panel (6/6) ruling: the S2 conjunctive gate failure (**not passed**) is upheld, and false alarms are recorded as non-discriminatory under §1.3. The stopping category is **Closed (negative result)** (limited to the frozen configuration: StormCast V1 + method (a) gust diagnostic + probability P≥0.3 / member median, 2 h). Residual attribution remains unseparated. Applying the HRRRCast heuristic to the same AI fields passes point estimates for BS, hit rate, and peak ratio, with false alarms at 16/120 versus 4/120 (report-only, post hoc). S4 will not run, and S3 will not launch. Testing the new definition requires a new pre-registration and a new blind sample (~180 compute units, CU), submitted as a recommendation to the parent node. Ruling recorded; finalization of the draft assigned.

10-27 21:25 lead verification: the false-alarm operator correction holds (deviation D7; consistent with pre-registration §4.3, with the diff changing only operators and paths). All 2640 merged successful records are individually traceable without duplicates or conflicts, and independent recalculations confirm all four point estimates. The S2 frozen conjunctive gate **failed** (BS, hit rate, and peak ratio all failed, independent of the correction). Under pre-registration §6, S4 will not run and S3 will not open automatically. The candidate stopping category is "Closed (negative result)" for this configuration; the six-reviewer panel is in progress, so no external conclusion is posted before the ruling. Decision recorded.

10-27 21:22 Eastern Daylight Time (EDT): Round 3 S2 groups 120–239 on Virtual Machine 2 (VM2) completed, with final checkpoints, logs, and `done` markers retrieved. Colab is stopped, with no active sessions in private or account lists. Merging Virtual Machine 1 (VM1) yields the frozen sample of 120 events plus 120 controls with 10 successful members each. Central processing unit (CPU) scoring shows BS, hit rate, and peak ratio failed while false alarms passed, so the S2 scaling conjunctive gate failed for this configuration. The round used ~176.4 CU (estimated from allocated duration × 6.77 CU/h, not an independent bill), below the approved 185 CU cap. The frozen scoring script mistakenly used a plus sign for the false-alarm difference. The initial output is preserved while a separately named corrected version uses a minus sign, awaiting lead review. Draft conclusion and audit recorded; results do not extrapolate to operational real-time forecasting or general AI versus HRRR superiority. Exploratory; not peer reviewed.

10-27 13:00 EDT: Round 3 S2 runs on the second virtual machine; scientific criteria remain unevaluated.

10-16 **On hold** (5/6 valid, ruling). What is on hold is sample expansion for the configuration "StormCast V1 / contiguous United States (CONUS) + 10 m wind × 1.6677 gust proxy + post-hoc fifth-generation European Center for Medium-Range Weather Forecasts reanalysis (ERA5) conditioning." Neither generation met the frozen expansion threshold (V1: 20 events; CONUS: 12 events, 53/54 groups). This finding **does not** mean convective-scale AI underperforms HRRR. Hit-rate differences are not significant (McNemar p 0.22–1.0), and Criterion 1 passes completely under the ensemble-maximum definition. Any-member counts reaching ≥25.7 m/s for AI match or exceed HRRR surface gust (GUST) (V1: 9/11/9 vs 8/8/6); V1 includes 7 seen cases and uses a constant gust factor. Expansion stops because the current design cannot determine whether scaling is worthwhile, so further sampling merely spends budget on non-significant differences (CONUS costs ~700 CU per 100 events). Any one of the following restart conditions triggers a new project proposal and newly pre-registered criteria rather than direct resumption of expansion: ① a blind sample with thunderstorm-wind verification, convective non-severe controls, pre-registered ensemble aggregation and probabilistic criteria, and a sample-size estimate; ② a mixed-layer wind-profile gust diagnostic replacing the constant factor; ③ native gust output, or a public convective-scale model testable outside its training period, becomes available; ④ a shift to batch uses that do not require station peaks (a goal change that the PI must decide). Round 2 totaled ~212 CU. Sub-project M is complete; sub-project L2 has not started.

Round 2 criteria (frozen 10-14, no post-hoc relaxation)

See the "Criteria" section: at each 1–3 h lead time, the median AI gust peak ratio must reach ≥0.9 times HRRR GUST. For ≥25.7 m/s detection (flagged when ensemble probability P≥0.3) across event plus non-event samples, the hit rate must at least match HRRR GUST. The false-alarm rate must not exceed 1.5 times HRRR GUST, and Global Forecast System (GFS) true-forecast conditioning must degrade by ≤20% relative to ERA5. The gust factor is fixed at 1.6677 (2022 fit). Reports list wide and narrow windows side by side, window-edge fractions, and two HRRR baselines; persistence appears only in the appendix.

Step 1 criteria (fixed before execution)

- Verification target: for each event, select the station recording the observed ASOS peak wind. Extract the model maximum 10 m wind (and gust variable, if available) within ±25 km of the station and ±1 h of the observed peak time. Baselines are operational HRRR forecasts at matching lead times (10 m wind and GUST) and persistence from the initialization-time HRRR analysis.
- Metrics: peak ratio (model peak / observed peak), peak timing and location biases, and binary detection of ≥25.7 m/s (50 kt).
- "Worth continuing" threshold: across 1–6 h lead times, the AI peak ratio must reach ≥90% of HRRR and beat persistence on at least 2/3 of events. Single-event single-initialization inference must also take ≤10 GPU minutes as a prerequisite for scaling. If the model falls short, document the exact gap without relaxing the criteria.
- Events must fall outside the training period of the weights used. StormCast V1 (trained 2018-07 to 2021-12, Apache-2.0 weights) uses post-2022 events; StormCast-CONUS (trained through 2025-09-30,

OpenMDW-1.1 weights) uses events after 2025-10.

Path to scaling up

Step 1 scripts run as a batch pipeline: event list → automated HRRR initialization retrieval → inference → automated ASOS verification. After passing the gate, the pipeline expands to dozens to hundreds of ASOS-measured thunderstorm wind events from the Storm Prediction Center (SPC) and Storm Events database for hindcast statistics. The AIWP statistics from sub-project M determine the predictable lead time of the environment, deciding whether sub-project L2 supports forecasting or only hindcasting. Cloud Model 1 (CM1) ensembles for sub-project L2 run on a supercomputer; scaling up is handed to the CM1 track or becomes a formal project.

Boundaries

- Hierarchy (root lead 10-14, 5-level cap): this node is a level-2 node and will not launch level-3 sub-projects. Subdivide work using tasks within this node, or ask the parent-node lead (Thunderstorms and Convective Scale) to launch a peer level-2 sub-project.
- Only two hops remain downward: this node → agent (ideas) → tools. Task briefs must not allow tool agents (gpu-gemini, reviewer, litwiki, etc.) to delegate further.
- Subtree lists are not grouped by parent node; the interface project tree defines the hierarchy.
- Compute and data: prioritize Colab without a single-trial cap (aligned on 10-14 by the parent project with the PI's instruction to use Colab extensively without CU limits, voiding the old ≤20 CU single-trial cap). Pause and report to the parent node if a single trial is projected to exceed 100 CU. Run `colab stop` immediately after completion to prevent idle sessions, and assign only one executor per round. Sub-project L Round 2 keeps its existing limits: ≤200 CU for the full round, reporting once at 120 CU cumulative and stopping at 200 CU. Supercomputer jobs are capped at ≤50 service units (SU) per job via formal resource requests; the CM1 track runs the sub-project L2 CM1 ensembles while we supply `input_sounding` profiles only.
- Compute and storage locations (tightened by the PI on 10-14): the workstation runs no compute and stores no compute data (46 GB free on drive C as of 10-15 01:59). Run and store all solvers, inference, training, batch cases, and large downloads on Colab, the Center for Research Computing (CRC), or the Texas Advanced Computing Center (TACC). Keep only scripts, small results under 200 MB per item, figures, and reports on the workstation; keep retrieved checkpoints small and leave large intermediates on Colab or the cluster. Report to the parent node before temporarily placing >1 GB on the workstation. Because CRC group storage is at ~95%, place new large datasets on TACC work or scratch storage first; CRC defaults to single-node jobs. Every task brief must specify where computations run and where outputs reside.

Decisions made

- 2026-10-14 literature review ruling: separate the two StormCast weight versions, and reclassify FuXi-Nowcast as a counterexample (underperforming persistence at severe-gust thresholds). Add HRRRCast as a candidate (public 6 km version, post-processed diagnostic GUST, pressure levels up to 200 hPa, GFS boundaries), and retract the claim that "only StormCast is usable." Narrow the scope of the Cook 10% finding, retract the Collins extrapolation, and frame novelty narrowly
- 2026-10-14 project launched: run Step 1 entirely on Colab and the workstation, and consult the CM1 track first on the CM1 component

Past trials and results

Date	Action taken	One-line result
10-16	Round 2 CONUS segment (Segment B 46.2 + Segment B run 2 (B2) minor + Segment B run 3 (B3) 78.1 CU, estimated): 12 events in 2026 + 6 controls, 53/54 groups (missing 1 member in 1 control 3 h group)	**Failed under frozen criteria**. Wide-window peak ratios at 1/2/3 h are 0.67/0.74/0.68 for AI/observed, 0.79/0.82/0.81 for HRRR GUST, and 0.84/0.94/0.86 for HRRR 10 m wind × factor. The AI/HRRR GUST ratio is 0.85/0.90/0.84, so Criterion 1 passes only at 2 h. Hits are 1/1/0 for AI versus 4/2/3 (/12) for HRRR GUST, failing Criterion 2; both have 0 false alarms, and the missing control group does not alter the conclusion. Exploratory; not peer reviewed.
10-15	Round 2 Segment A: V1 + ERA5 conditioning full sample of 20 events + 10 non-event controls × 1/2/3 h × 10 members (~25.7 CU this session, rate-estimated)	**Failed under frozen criteria**. Wide-window peak ratios (AI / HRRR GUST / HRRR 10 m wind × factor) are 0.68/0.73/0.90 at 1 h, 0.68/0.71/0.75 at 2 h, and 0.64/0.75/0.85 at 3 h. Criterion 1 (AI ≥0.9 × HRRR GUST) passes at 1 h (0.93) and 2 h (0.96) but fails at 3 h (0.85). Criterion 2 fails with AI hits at 4/5/4 versus HRRR GUST at 8/8/6 (/20), while control false alarms are 0/10 for both. Factor-scaled HRRR 10 m wind also mostly outperforms AI; narrow-window ratios reach only 0.4–0.5 for both, with 133–143/300 AI members on the window edge. Exploratory; not peer reviewed.

Date	Action taken	One-line result
10-15	Round 2 gate segment: V1 ran the gust pipeline on the 7 Step 1 events (~11.7 CU this session, estimated from rate × duration)	Pipeline and runtime passed the gate: 173–312 s per group, member median 22 s (projected 55 s), costing ~0.3–0.6 CU/group. On the 7 already-seen events (not a blind test), wide-window 1–3 h peak ratios are 0.51/0.59/0.62 for AI/observed and 0.75/0.73/0.77 for HRRR GUST, with AI hits at 0/3, 0/3, and 0/2 of HRRR GUST hits. Across 10 pooled events, median AI/HRRR GUST ratios are 0.84 at 1 h, 0.88 at 2 h, and 1.01 at 3 h; ≥25.7 m/s hits are 3/2/2 for AI versus 6/6/4 for HRRR GUST, while both miss all 7 cases in the narrow window. Exploratory; not peer reviewed.
10-14	Round 2 Segment ② V1/ERA5 (old cap ≤20 CU counted in this round; upper bound 18.03 CU from account balance drop)	Saved 153 checkpoint rows and 13 complete 10-member groups: 3 events (AMA, CDS, CID) × 1/2/3 h, plus 4 control groups (CDS at 3 lead times, CID at 1 h). Nine of 10 controls were cross-checked against neighboring METAR stations (<40 kt), while CNK has no neighboring station. Interim medians are 0.79 at 1 h, 0.83 at 2 h, and 0.77 at 3 h for AI/observed versus 0.91/0.92/0.79 for HRRR GUST; ≥25.7 m/s hits are 3/3, 2/3, 2/3 for AI and 3/3, 3/3, 2/3 for HRRR GUST, with 0 control false alarms. About 7 new groups used ≤18 CU, projecting >200 CU for the full ≥20 + ≥10 sample pending gate timing verification. Exploratory; not peer reviewed.
10-14	Round 2 Segment ① HRRRCast GPU compatibility (~1.1 CU)	Weight checksums passed again, and isolated Python 3.10 + TensorFlow 2.15.1 recognized the A100 GPU, but loading `.keras` hung the kernel and blocked event inference. Decision: put HRRRCast on hold (public 6 km resolution, training covers 2023, environment blocked) and retry with post-2024-04 events only if the V1 segment passes. Exploratory; not peer reviewed.

Date	Action taken	One-line result
10-14	Round 2 partial execution (pilot)	Across two V1 cases (AMA 65 kt, CDS 62 kt) × 1/2/3 h × 10 members, the gust proxy (AI 10 m wind × 1.6677, 2022 fit) gave wide-window AI/observed ratios of 0.77–1.04, comparable to HRRR GUST at 0.72–0.92, with P(≥25.7) at 1.0 for AMA and 0.3–0.7 for CDS. CONUS ran on an A100 80 GB GPU (forward pass 97 s / 2 steps, 20.6 GB), yielding an AI/observed ratio of 0.66 and P=0 at L1 for one Oklahoma City (OKC) 2026 case, with CONUS pressure fields reaching only p20hl (~440 hPa). Single-member GFS conditioning passed a smoke test, whereas HRRRCast weights passed verification (trained 2021-03 to 2024-03, covering 2023 events) but preprocessing failed before inference; the platform pruned the first session, leaving all three criteria unevaluated. Exploratory; not peer reviewed.
10-14	HRRRCast availability check	Public weights exist (~203 MB), but the public release uses 6 km resolution, reaches only 200 hPa, diagnoses GUST via post-processing (factor 1.4), lacks a standalone weight license, and has an unverified training period; not yet tested on Colab. Exploratory; not peer reviewed.
10-14	Step 1 observation and baseline supplement: case-by-case peak check using Iowa Environmental Mesonet (IEM) 1-minute ASOS data; weight check in the WoFSCast repository	In 6/7 cases, 1-minute data match the reported peak in both minute and value; OFK lacks data for that minute (window maximum 53 kt, METAR peak wind (PK WND) 56 kt) and is listed separately, though same-source comparison cannot rule out in-station quality-control peak removal. Hits ≥25.7 m/s are 0/28 for AI, 8/28 for HRRR GUST, and 1/28 for HRRR 10 m wind. The WoFSCast repository lacks weights or releases and points to internal paths, making it unavailable and leaving the Round 1 failure unchanged. Exploratory; not peer reviewed.

Date	Action taken	One-line result
10-14	Sub-project M: Amazon Web Services (AWS) AIWP models (Pangu, GraphCast Operational) across 40 ASOS ≥50 kt thunderstorm wind days in 2023–24 at 12–120 h lead times against 13-level ERA5, using diurnal-cycle-aligned persistence and per-parameter criteria	Downdraft convective available potential energy (DCAPE) and 0–1 km wind shear pass all five lead times; 0–6 km shear passes all five for Pangu and through 72 h for GraphCast. Both models fail all five lead times for most-unstable and mixed-layer convective available potential energy (MUCAPE/MLCAPE), beating persistence on only 14–25/40 events (under 2/3) even though median errors are often below persistence. Verification uses the nearest 00/12 UTC time, which can be up to 6 h from the wind peak, and evaluates environments rather than gusts. Exploratory; not peer reviewed.
10-14	Step 1: StormCast V1 (trained 2018-07 to 2021-12, central United States 512×640 grid) across 7 ASOS thunderstorm wind events in 2023–24 × 1/2/3/6 h lead times	Failed preset thresholds: median AI/HRRR 10 m wind peak ratios are 0.85 at 1 h and 0.61 at 2 h (1.11 at 3 h, 1.01 at 6 h), with 0/28 AI runs detecting ≥25.7 m/s and 6/7 events beating persistence (≤88 s/group, ~6 GPU hours for 100 events × 4 initializations). Both AI and HRRR 10 m wind represent 10 m mean wind, reaching only 0.3–0.5 of ASOS gusts versus 0.75 for HRRR GUST, using single members and post-hoc ERA5 conditioning. Soundings cover only 14 model levels to ~440 hPa, making convective available potential energy (CAPE) unusable for sub-project L2; WoFSCast and FuXi-Nowcast lack public weights, and CorrDiff weights are restricted. Exploratory; not peer reviewed.
10-14	Deep Research verification	Both reports agree that no convective-scale AI model has been verified against stations for surface thunderstorm severe winds (none found in the search), and that in practice StormCast is the only usable model. Deep Research misattributed the HRRR severe-wind bias to Fovell 2022 rather than Collins 2024 in Weather and Forecasting (WAF) PDF p10 (gust bias −11.32 m/s in cold-season downslope winds). ASOS quality control erroneously removes ~10% of convective peaks (source PDF p1), and no "AI sounding → CM1" precedent exists; sent for review. Exploratory; not peer reviewed.

Date	Action taken	One-line result
10-14	Grok literature scan + verification of the StormCast model card and FuXi-Nowcast abstract	Current StormCast weights cover 2018-07–2025-09 across the full CONUS at 3 km under the commercial-permissive OpenMDW-1.1 license with 10 m wind but no gusts, requiring verification events after 2025-10. FuXi-Nowcast includes gusts but covers only East China without public weights, and no "AI sounding → CM1" precedent was found. Exploratory; not peer reviewed.

Parameter sources

Parameter	Value	Source or "assumption"	Sources checked
Verification neighborhood ±25 km / ±1 h	25 km, 1 h	Assumption: ~8 grid points at 3 km covering convective position error; sensitivity: also report ±10 km / ±0.5 h	Pending rapid literature scan
Severe wind threshold	25.7 m/s (50 kt)	National Weather Service (NWS) severe wind standard	—

Open items and next steps

- 10-22 03:30 VM1 disk filled up, halting S2 at 120/240 (60 events + 60 paired controls; all 10 members of unit 120 require rerunning). The run consumed ~91.3 CU; full-round extrapolation reaches 180.1 CU at the effective rate and 182.1 CU at the conservative rate, exceeding 180 CU and preventing VM2 launch. We reported to the parent node and requested a cap increase to 185 CU (corrected). Once approved, VM2 will fix the disk leak before resuming. Logged.
- 10-21 19:32 lead re-estimation: ~37.2–37.5 CU used with units 20–47 completed; extrapolation reaches 175.7–177.6 CU, or 177.6–179.5 CU including VM2 overhead, remaining within 180 CU. Agent wx's follow-up slots are taken by two expired follow-ups, so I will dispatch VM2 directly at the 10-22 04:26 follow-up.
- 10-21 19:30 S2 continuation: VM1 (started at 16:28) runs units 20–129 and reached ~unit 46 by 19:23. The first 10 groups measured 6.77 CU/h (0.721 CU/group), extrapolating to 175.6–177.5 CU before second-machine overhead and leaving only ~2.5–4.4 CU below the 180 CU cap. The original `--keep 5` keepalive setting deleted 2 early non-final checkpoints and was updated to `--keep 1000`. Agent wx was assigned to re-estimate every 40 groups, launch VM2 for units 130–239 after VM1 finishes, score the outputs, and draft the conclusion. I scheduled a fallback callback for 10-22 04:26.

- 10-21 13:11 S2 first 20 groups completed (20/20 groups, 10/10 members; 387.7 s/group, or 0.738 CU/group at 6.85 CU/h). Full-round extrapolation reached ~178.9 CU, exceeding 160 CU, so agent wx stopped and reported. The PI approved raising the cap to 180 CU. At 13:3x, agent wx was dispatched to resume groups 20–239 across two sequential virtual machines under deviation D6 (~23.7 h remaining, exceeding the 24 h single-machine limit), reporting both measured and conservative rates.
- 10-21 The PI approved releasing S2 (Colab ≤160 CU), and the platform registered the gate. At 09:5x, agent wx was assigned the S2 task brief and queued due to full gateway delegation slots. After completing the first 20 groups, agent wx will report the session name, per-group CU, and full-round re-estimation.
- 10-16 01:52 Round 3 pre-registration frozen:
- Finalized with SHA256 recorded.
- Based on the review ruling and decision record.
- Dispatched agent wx to execute S0 + S1 (Colab, ≤12 CU).
- 10-16 02:10 agent wx returned pre-screening (~1.7 CU): 295 provisional V1 candidates and 127 CONUS candidates (pending grid-boundary checks and control pairing, not yet frozen); S1 had not started. At 02:20, dispatched a follow-up to finish S0 and proceed to S1 within ≤12 CU combined.
- 10-16 02:28 return: the two S0 jobs were estimated at ~3.4 CU at the A100 rate, exceeding the 3 CU segment estimate, so agent wx stopped. The sample remained unfrozen and S1 had not started (grid checks confirmed all provisional candidates lay inside the domain). Recorded as deviation D1: segment CU values are budget estimates, raising S0 to ≤5 CU while keeping S0 + S1 at ≤12 CU; dispatched at 02:36 to complete both.
- 10-16 03:26 return halted on a billing issue: the CPU session was billed at the A100 rate of 6.85 CU/h and stopped at an estimated ~9.8 CU. Qualified event pools reached 301 for V1 and 132 for CONUS, with 120/80 provisionally sampled and control pairing incomplete. Recorded as deviation D2 to bill CPU sessions at the measured rate, and dispatched at 03:36 to finish control pairing, sample freezing, and S1.
- 10-16 04:56 check: control pairing ran in the background on the CPU session at 22/200 with no missing controls; measured CPU usage was ~0.08 CU/h, confirming deviation D2. Agent wx scheduled a 07:45 callback to freeze the sample and run S1, A note was left requiring S1(a) to use all native HRRR wrfnat levels.
- 10-16 10:49 S0 frozen (event and control lists): 120 V1 events, 80 CONUS events, and all 200 controls paired (qualified pools: 301 V1, 132 CONUS; switching to 35 dBZ added 0 events). At 11:13, initial S1 results showed (a) 1.113 / r=0.915, (b) 1.00, and (c) failed (AI Z1hl minus terrain ranged from −26 to 39 m). A lead review found that the virtual potential temperature (θv) calculation treated Pa as hPa, voiding the initial run as deviation D3, while height was anchored to the lowest HRRR level as deviation D4; agent wx was dispatched at 11:32 to rerun S1 (~4.3 CU cumulative).
- 10-16 11:38 corrected S1 **passed all checks**: (a) formula versus operational 10 m gust (`fg10m`) median ratio 0.99999, r=0.982 (10 events, 4370 grid-point × time steps); (b) median peak ratio 1.000 using only the 14 StormCast levels; (c) 0 NaN values and planetary boundary layer (PBL) tops found across all 860 columns in two events after anchoring. AI Z1hl drift (−26 to 39 m) was anchored under deviation D4, and non-monotonic height columns were handled under deviation D5, totaling ~6.4 CU; reported to the parent node, **awaiting parent-project approval for S2 (V1 120+120, ~144 CU)**.

- (Original plan) Report to the parent node after passing the S1 gate. **S2 awaits parent-project approval and does not launch directly.**
- 10-16 The parent node "AI Meteorology" approved the direction and released S0 + S1 (~9 CU, contingent on post-review finalization). Dispatched agent litwiki to build literature cards (Benjamin 2020 gust diagnostic text and radar convective thresholds) and sent the final draft for a review by three independent reviewer models, examining V1 80+80 statistical power and whether to raise the sample above 100 if more than 80 events qualify. Once both return, we will adjudicate, freeze the draft, and release S0 + S1.
- Agent litwiki returned on 10-16; spot-checking pages 14 and 16 of the Benjamin 2020 PDF confirmed accuracy, requiring two final-draft revisions: 1. Set the mixed-layer top criterion to θv(hl1, ~8 m) + 0.5 K (Benjamin 2020 PDF page 14 / report page 10), taking the maximum only within the PBL (PDF page 16 / report page 12). This eliminates the 2 m specific humidity (`q2m`) assumption. 2. Keep the 40 dBZ radar threshold as an "assumption": the literature supports only Multi-Radar Multi-Sensor (MRMS) composite reflectivity >40 dBZ (Liu 2021, technical note page 2), while Roberts & Rutledge 2003 >35 dBZ (PDF page 1 / journal page 562) serves as a sensitivity check. Report to the parent node once S1 passes; **do not launch S2 directly** before parent-project release.
- 10-16 Round 3 pre-registration draft completed (**design only, zero compute spent**). Sub-project L remains On hold pending release by the parent node "AI Meteorology." Draft highlights:
- Blind sample: V1 uses events from 2022, 2024, and 2025-01–09, while CONUS uses national events after 2025-10; controls have convection without severe wind, paired 1:1 with events.
- Gusts switch to the HRRR mixed-layer diagnostic (Collins 2024 Equation (1), journal page 708), requiring reruns that save levels hl1–hl20.
- Primary lead time is 2 h with three primary criteria: non-inferior Brier score, non-inferior hit and false-alarm rates (margin 0.15), and peak ratio ≥0.9.
- Sample size: 80 events + 80 controls for V1, and ≥50 for CONUS.
- Stages: S0 sampling freeze + S1 pilot + S2 V1 blind test take ~105 CU; all five stages take ~570 CU, with S2 onward requiring prior parent-node approval.
- Pre-finalization: assign the Benjamin 2020 text and radar thresholds to agent litwiki for card creation, then submit the draft for review under the freeze protocol.
- 10-15 12:50 Segment B session stopped by the watchdog at 11:25 (46.19 CU, 230 rows, 20 complete groups); the new Segment B run 2 (B2) session exited immediately with `OptionalDependencyError` and produced no results (reported to the platform). At 12:58, dispatched agent wx to launch Segment B run 3 (B3) with all optional dependencies installed to resume from row 230 to 54 groups, with ~81 CU remaining in the round (~134 CU cumulative) before stopping to report.
- 10-15 Parent-project approval: run CONUS Segment B to all 54 groups with a full-round cap of 215 CU, replacing the 200 CU stop limit; resume within the same Segment B without recomputing or opening duplicate sessions, and stop to report if exceeding 215 CU. Notified agent ideas: if the run fails, send it for a review by six independent reviewer models before putting it on hold; if it passes, report directly to the parent node. Scaling costs (~700 CU for CONUS, ~140 CU for V1) are logged for reference only and do not authorize release.
- 10-15 10:42 Recalculation from actual Segment B runtime (requested by the parent node): the Segment B session ran ~6.0 h and used ~41 CU (estimated at 6.85 CU/h), completing 17/54 groups. Wall-clock time per group (1 event × 1 lead time × 10 members) is ~14 min for 1 h, ~20.5 min for 2 h, and ~27 min

for 3 h (averaging ~20.5 min ≈ 2.3 CU/group, ~1.6 times slower than the 13 min/group estimate at 05:20 based on two 1 h groups). Round 2 cumulative usage reached ~83 + 41 ≈ 124 CU, passing the 120 CU reporting threshold (reported to the parent node). The remaining 37 groups require ~87 CU, projecting ~210 CU for the full round and hitting the 200 CU cap. At 200 CU, we will stop and evaluate the ~49/54 completed groups under the frozen criteria while documenting sample counts (prioritizing events over controls), correcting the 05:20 projection of ~175 CU.

- Direct scaling cost evidence: on a single A100 GPU, CONUS takes ~20 min/group, so 100 events × 3 lead times = 300 groups ≈ 100 GPU hours ≈ 700 CU; V1 takes 3–5 min/group, requiring ~20 GPU hours ≈ 140 CU at the same scale.
- 10-15 05:20 Round 2 accounting audit (requested by the parent lead):
- By segment (in CU; "upper bound" = shared account balance drop including concurrent sessions; "estimated" = session duration × ~6.85 CU/h): pilot ≤26.8 (upper bound); Segment ① HRRRCast ~1.1 (estimated); Segment ② V1 ≤18.0 (upper bound); gate segment ~11.7 (estimated); Segment A ~25.7 (estimated); Segment B through 05:10 ~3.3 (estimated), totaling ~86.6 CU.
- Accounting basis: the previous project-page figure of "upper bound ~58" covered usage before Segment A (≤46 + 11.7), while the reported "~84/87" equals 58 + Segment A 25.7 (+ Segment B 3.3). No double-counting or omission occurred between the two timestamps, and the upper bounds for the pilot and Segment ② overstate actual usage.
- Segment B projection: at ~13 min/group and ~1.65 CU/group (the first 2 groups included installation and weight loading), 54 groups take ~89 CU, with 3.3 CU used by 05:10, about 86 CU remain, plus ~2–3 CU for installation and loading in one resumed session. The full round was projected at ~175 CU, passing 120 CU (to be reported) without hitting 200 CU; this retracts the erroneous 05:10 claim that it would likely hit 200 CU.
- The 10:40 callback will recalculate this projection using actual per-group runtime from Segment B.
- 10-15 05:10 CONUS segment (Round 2 Segment B) in progress: completed 2/54 groups in ~27 min, running much slower per group than V1. The parent lead approved opening a new session from checkpoints at 50 CU per session with a ≤200 CU round cap and a report at 120 CU. If unfinished at 200 CU, evaluate completed groups under the frozen criteria while reporting sample counts, per-group GPU minutes and CU, and estimated cost for 100 events × 3 lead times.
- 10-15: Round 2 files >100 KB (including the gate-segment checkpoints) were moved to TACC archive storage during the root lead's cleanup (with a manifest); the root lead will verify and restore them after TACC recovers. Resumed runs recover only from Colab, noting "workstation checkpoints on TACC, not compared" if a ruling depends solely on those files (reported to the platform).
- Round 2 cumulative Colab usage (upper bound, including concurrent shared-account sessions): pilot ≤26.8 + Segment ① ~1.1 + Segment ② ≤18.0 ≈ ≤46 CU, with a parent-node report due at 120 CU. The gate segment passed on 10-15 at 00:26, bringing the cumulative upper bound to ~46 + 11.7 ≈ 58 CU. Released Segment A (V1 padded to ≥20 events + ≥10 controls, ≤50 CU) and Segment B (CONUS 2026 events + ≥6 controls, ≤50 CU), holding the GFS segment until a model passes Criteria 1 and 2.
- Agent ideas Round 2 (task brief queued after sub-project M): covers gust conversion, 10-member ensembles, ≥20 events + ≥10 non-event controls, GFS true-forecast conditioning, and StormCast-CONUS trials on post-2025-10 events with vertical-level checks under fixed criteria. The 10-14 18:20 pre-check estimated ~163 CU across 666 candidates and paused at the 100 CU gate. After the lead initially released 200 CU, the parent lead revised the cap on 10-14 to ≤20 CU per segment (pending parent-project

approval for 200 CU), dropping the 6 h lead time and batching ensemble members in the order HRRRCast → V1 ERA5 → GFS conditioning → CONUS 2026 events.

- Parent lead 10-14 added two diagnostic readouts (forwarded to agent ideas without altering criteria): report the fraction of maxima falling near the window edge (≥0.9 × radius) and list narrow and wide windows side by side, since ~20/28 Round 1 AI wide-window maxima lay at 20–25 km and reflected background wind. Because the persistence baseline is too weak for "6/7 beating persistence" to prove skill, conclusions rely on the HRRR GUST and "HRRR 10 m wind × same gust factor" baselines. Round 1 re-verification showed wide-window maxima near the edge (≥22.5 km) in 18/28 AI, 13/28 HRRR 10 m, and 14/28 HRRR GUST runs, while narrow-window (10 km / ±0.5 h) detections ≥25.7 m/s were 0/28 for all three.
- Sub-project L2 soundings: StormCast V1 reaches only ~440 hPa and cannot drive CM1 directly; candidate alternatives are HRRR pressure-level products or the CONUS version (pending Round 2 level verification).
- 10-14 sounding supplement: CM1 column-extraction rules were coded into a testable script (passing 2/2 synthetic tests), while radiosonde observations (RAOB) across the seven cases differed from ASOS peak times by 0.25–4.35 h (3 cases >3 h). True pre-convective CM1 columns remain unextracted because they require the first radar ≥35 dBZ timestamp, HRRR reflectivity, and land masks per case, and raw profiles were not saved; `soundings.md` serves as reference only rather than a sub-project L2 handoff. Decision: defer sub-project L2 column extraction and `input_sounding` generation (using native HRRR/HRRRCast levels + Next Generation Weather Radar (NEXRAD)[local path omitted]) until Round 2 proves "worth continuing."
- Sub-project M completed (see table above): global AI models skillfully predict wind shear and DCAPE 3–5 days ahead, whereas CAPE remains unstable across 0.5–5 days. For sub-project L2, hindcasts can extract soundings from HRRR or ERA5 analyses, whereas forecasts remain bottlenecked by CAPE and must rely on short lead times from convective-scale models (Round 2 HRRRCast includes CAPE and 20 levels); no further work will be added to sub-project M.
- Grok scan and Deep Research verified, with review adjudicated; HRRRCast was added to Round 2, and agent ideas will provide the GitHub repository table in `models.md`.
- 10-14 CM1 track reply: no duplication (they extract columns only from HRRR f00 analyses), and their extraction rules were shared with agent ideas; after 2022, only NIP (2022-06-03) and ALTU (2023-07-03) have measured peaks, both inside the current StormCast training period and excluded from verification. The CM1 track will run the small sub-project L2 ensemble (10–20 members) using `input_sounding` profiles we supply, scheduled after CRC group storage is restored and after their sub-projects C and B launch.

Literature basis

- CM1 downburst ensembles driven by 97 soundings (sub-project L2 precedent; PDF p3)
- Near-domain precedents (cards not yet built): CGsNet Xiao et al. 2023 Geoscientific Model Development (GMD) 10.5194/gmd-16-3611-2023; Lagerquist et al. 2017 WAF 10.1175/WAF-D-17-0038.1; FuXi-Nowcast arXiv 2512.08974v2 (station gust verification, underperforming persistence at 13.9/17.2 m/s); HRRRCast

arXiv 2507.05658; Cook 2023 Meteorology 10.3390/meteorology2020017; Collins et al. 2024 WAF PDF page 10
- Bolgiani 2020
- Parent project AI Meteorology judgment table row 1b: StormCast arXiv 2408.10958, WoFSCast 10.1029/2024GL112383, FuXi-Nowcast arXiv 2512.08974, Das 2024 10.1038/s41612-024-00834-8 (cards not yet built)

Note from the PI

- 10-14: Once models reach the 3 km scale and capture favorable conditions, consider downscaling with CM1; precipitation is harder to predict and remains a future hazard candidate (sub-project P, not launched for now). The ultimate goal is scalable forecasting and hindcasting; use Colab extensively at this stage without CU limits.

10-22 citation snowballing (the PI requested re-running across all sub-projects)

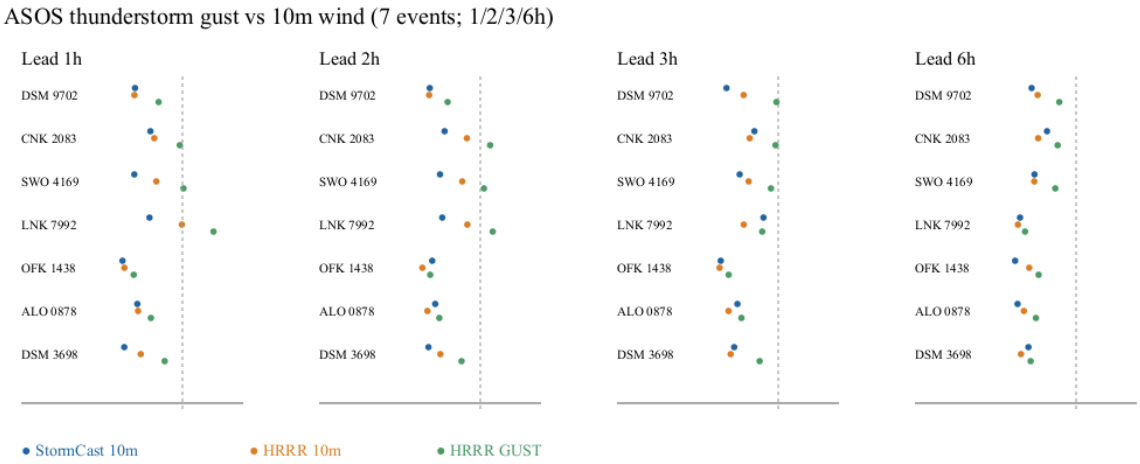
- Pool (4625 papers, 1733 after initial screening, 884 from snowballing alone; includes Semantic Scholar). Selected 13 papers for agent litwiki batch 7 (pending submission, automatic background retry).
- Notable at the title level: AIWP models directly outputting severe convective forecasts (aies-d-25-0065.1), AI forecasts plus Transformer post-processing for medium-range severe weather (aies-d-25-0045.1), and case comparisons of AI models on storm impacts (arXiv:2511.01665). These may supplement literature comparisons on "AI convective background fields reaching only 12–24 h," but conclusions remain unchanged until literature cards and full texts are verified.

10-22 batch 7 verification

- **Narrow the conclusion scope**: the finding that "global AI convective background fields reach only 12–24 h" applies only to the three events in this track evaluated via local environmental variables (such as CAPE). White 2026 (Artificial Intelligence for the Earth Systems (AIES), 10.1175/aies-d-25-0065.1) combines Pangu environmental fields with random forests to predict 1–8 day severe convective hazard probabilities (wind, hail, tornado) in 2024 tests, achieving Brier skill score (BSS) >0 across all lead times and matching the Global Ensemble Forecast System multilayer perceptron (GEFS-MLP) through day 7 (published PDF p.7, spot-checked by the lead); Hua 2026 reports similar 3–8 day skill. Both predict regional hazard occurrence probabilities rather than local environmental variables or station gusts, so they are not same-metric counterexamples, but they show that "AI background fields + statistical post-processing" holds skill at medium range.
- Other judgments in this track (StormCast configuration falling short, v2 scoped gap) remain unchanged; ProbSevere v3, Jahan 2024 (Weather Research and Forecasting (WRF) + machine learning (ML) gusts),

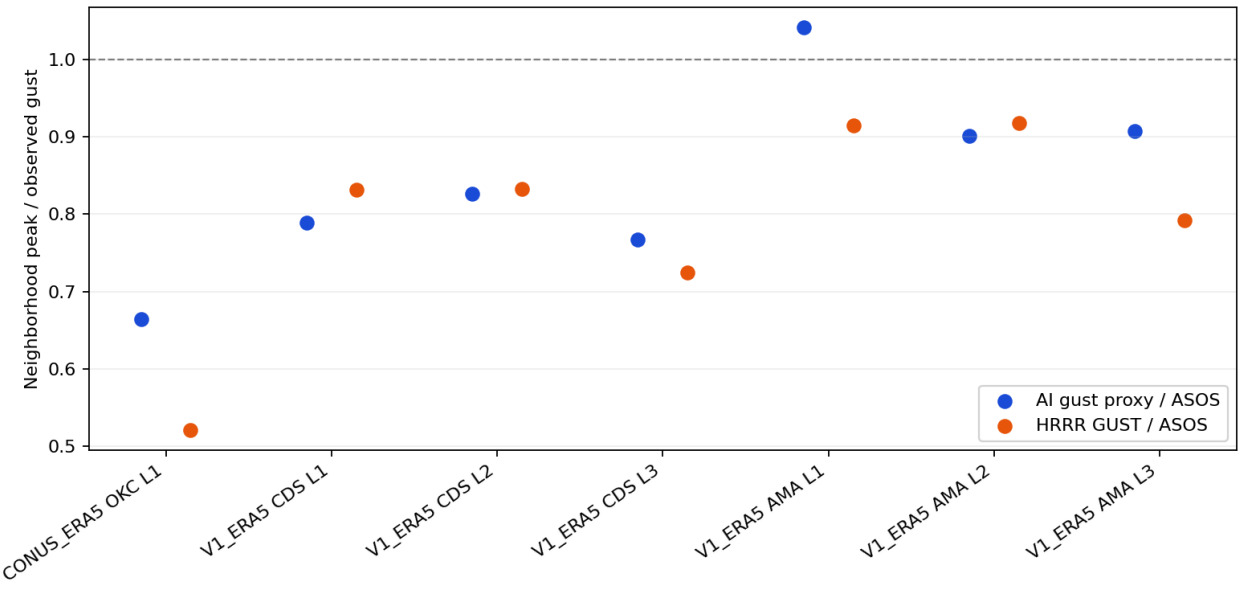
and Hallgren 2024 (lidar-trained local hub-height wind) serve as adjacent precedents.

Key figures



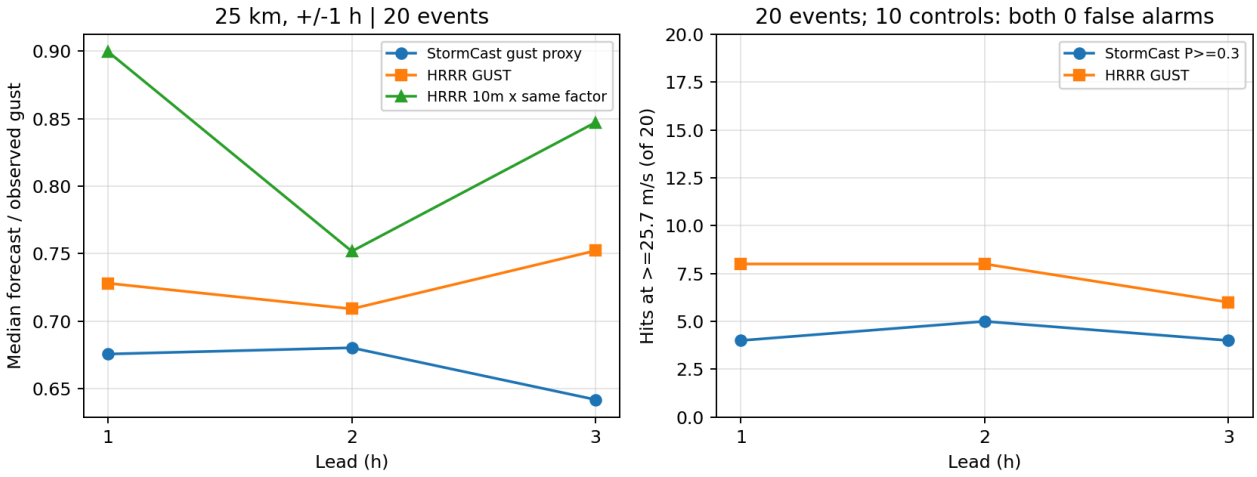
Sources: NOAA Storm Events; IEM ASOS; NOAA AWS HRRR; Copernicus ERA5/ARCO; NVIDIA StormCast V1. Dashed line = forecast/observed 1.0.

peak ratio



Pilot only (2 V1 events + 1 CONUS event); NOAA Storm Events/IEM ASOS, AWS HRRR, ARCO ERA5, NVIDIA StormCast. Not r2 gate.

pilot peak ratio



Sources: NOAA HRRR / Storm Events, IEM ASOS; NVIDIA StormCast V1; ARCO-ERA5.
Gust proxy = 10 m speed x 1.6677 (2022 fit); retrospective ERA5, not operational.

v1 verdict